

YARKOVO, Russian Federation

WMO: 283660

Lat: 57.417N

Lon: 67.083E

Elev: 171

StdP: 14.61

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.5	(c) -22.4	(d) -32.4	(e) 0.9	(f) -26.5	(g) -27.6	(h) 1.2	(i) -22.0	(j) 14.4	(k) 11.4	(l) 12.8	(m) 10.4	(n) 2.5	(o) 340	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	84.9	67.6	81.9	66.6	78.9	65.4	70.2	81.3	68.4	78.9	66.9	76.3	5.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.1	75.2	64.6	91.7	73.5	63.2	87.3	72.2	34.2	81.6	32.8	79.1	31.6	76.6	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.3	12.2	10.9		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	-33.8	72.4	4.5	2.6	-37.1	74.3	-39.7	75.8	-42.2	77.3	-45.5	79.1	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.8	2.5	8.7	22.0	40.3	51.9	62.8	64.9	61.6	49.7	35.3	19.6	8.3	(6)
(7)		DBStd	24.25	13.32	13.32	10.43	8.65	9.62	7.96	6.50	7.28	7.33	9.41	12.94	14.02	(7)
(8)		HDD50	6672	1472	1157	868	313	95	5	1	5	91	462	911	1292	(8)
(9)		HDD65	10879	1937	1577	1333	742	418	136	83	156	459	920	1361	1757	(9)
(10)		CDD50	1479	0	0	0	21	155	388	463	363	83	6	0	0	(10)
(11)		CDD65	213	0	0	0	0	13	69	81	49	1	0	0	0	(11)
(12)		CDH74	2002	0	0	0	1	183	663	748	400	7	0	0	0	(12)
(13)		CDH80	461	0	0	0	0	32	177	167	85	0	0	0	0	(13)
(14)	Wind	WSAvg	5.1	4.9	4.9	6.0	6.1	6.0	4.9	4.4	4.4	4.5	5.4	5.3	4.7	(14)
(15)	Precipitation	PrecAvg	21.9	1.4	1.0	1.4	1.1	1.8	2.4	3.0	2.4	1.9	1.9	2.0	1.7	(15)
(16)		PrecMax	26.9	2.8	2.2	2.8	2.2	3.7	5.0	4.6	4.2	3.3	4.4	4.2	3.5	(16)
(17)		PrecMin	16.7	0.1	0.1	0.1	0.1	0.5	0.7	0.9	0.8	0.4	0.2	0.4	0.6	(17)
(18)		PrecStd	2.8	0.7	0.5	0.7	0.6	0.8	0.9	1.1	0.8	0.7	1.0	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	38.4	45.0	70.0	84.3	88.2	88.9	86.4	74.7	65.3	42.8	34.7	(19)
(20)			MCWB	29.3	33.6	37.0	55.2	59.8	69.4	69.5	69.8	60.9	51.0	40.2	32.0	(20)
(21)		2%	DB	28.7	35.0	40.5	65.3	79.2	84.7	84.3	82.3	70.5	56.3	39.6	31.7	(21)
(22)			MCWB	27.2	32.1	34.6	52.4	61.0	67.8	68.0	68.9	58.6	47.3	37.7	30.6	(22)
(23)		5%	DB	25.6	31.6	38.3	60.0	74.8	81.5	81.1	78.6	65.4	51.1	36.7	28.9	(23)
(24)			MCWB	24.3	29.1	33.5	49.8	57.7	66.8	66.7	67.1	55.8	45.8	35.0	27.6	(24)
(25)		10%	DB	21.7	27.7	35.9	55.2	70.0	77.5	78.4	74.8	61.7	48.0	34.4	25.9	(25)
(26)			MCWB	20.5	26.0	32.1	46.8	55.6	64.6	65.6	64.9	54.2	44.4	32.7	24.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.6	34.4	37.9	57.5	65.1	73.4	71.8	72.0	62.9	52.4	41.1	32.7	(27)
(28)			MCD B	30.5	37.6	42.3	67.9	77.2	87.0	83.1	82.5	71.5	62.0	42.3	33.7	(28)
(29)		2%	WB	27.5	32.2	35.7	54.1	61.9	69.8	69.9	69.5	59.7	49.3	37.7	30.5	(29)
(30)			MCD B	28.8	35.3	39.7	62.4	74.8	81.3	81.6	79.7	68.5	53.6	39.0	31.7	(30)
(31)		5%	WB	24.3	29.1	34.1	50.8	59.2	67.7	68.3	67.8	57.0	46.7	35.2	27.7	(31)
(32)			MCD B	25.6	31.3	37.3	59.2	72.3	78.7	78.7	77.0	63.5	50.7	36.5	28.8	(32)
(33)		10%	WB	20.7	25.9	32.5	46.8	56.6	65.7	66.9	66.1	54.7	44.0	32.8	24.7	(33)
(34)			MCD B	21.7	27.7	35.8	54.8	68.1	75.5	75.9	73.9	60.4	48.0	34.1	25.9	(34)
(35)	Mean Daily Temperature Range	MDBR		13.2	14.5	16.6	17.0	20.6	19.3	18.3	18.8	15.3	11.6	10.9	11.7	(35)
(36)		5% DB	MCD BR	12.6	13.3	16.0	23.7	26.4	24.6	23.7	23.3	20.9	17.6	10.6	10.5	(36)
(37)			MCWBR	12.3	11.3	11.5	13.9	12.0	10.6	9.8	11.5	11.1	11.3	9.5	9.9	(37)
(38)			MCD BR	12.3	12.8	14.0	21.7	24.3	21.8	20.5	21.3	19.2	16.9	9.5	10.7	(38)
(39)		5% WB	MCWBR	12.0	11.1	10.8	13.2	12.5	10.1	9.6	10.8	10.5	11.8	8.8	10.2	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.279	0.311	0.368	0.412	0.407	0.408	0.419	0.427	0.379	0.349	0.306	0.278	(40)
(41)		taud		2.026	2.028	1.943	2.072	2.242	2.307	2.280	2.221	2.318	2.254	2.106	1.983	(41)
(42)		Ebn at Noon		173	221	238	251	263	264	258	245	240	210	161	132	(42)
(43)		Edh at Noon		19	30	43	44	40	38	38	38	30	24	17	15	(43)
(44)	All-Sky Solar Radiation	RadAvg		219	529	1001	1405	1658	1809	1721	1330	836	418	196	127	(44)
(45)		RadStd		17	37	64	113	114	128	158	126	89	64	30	18	(45)

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page